1. Vectors in $\mathbb{R}^n$

operations

addition: $u+v=(u_1+v_1,\dots,u_n+v_n)$

scalar multiplication: $c\cdot u=(c\cdot u_1,\dots,cu_n)$

magnitude

$\|v\|=\sqrt{v_1^2+\dots+v_n^2}$

distance

$d(u,v)=\|u-v\|$

Dot product

$u\cdot v=\sum u_i\cdot v_i$

use for: angles, projections, orthogonality

orthogonal

$u\cdot v=0$

angle

$\cos\theta=\dfrac{u\cdot v}{\|u\|\cdot\|v\|}$

Projection

$\text{proj}_v(u)=\dfrac{u\cdot v}{\|v\|^2}\cdot v$ → projection of u onto v

triangle equality

$\|u+v\|\le\|u\|+\|v\|$

3. Vector spaces

Vector space axioms ⤷ rule for a set of vectors (V) $\vec{v}$ to be a vector space ⑩

Closure:

① $\vec{u},\vec{v}\in V\to\vec{u}+\vec{v}\in V$

② $\vec{u}\in V$ & $c\in\mathbb{R}\to c\vec{u}\in V$

existence:

③ $\exists\ \vec{0}\in V$ s.t $\vec{0}+\vec{v}=\vec{v}$ for every vector $\vec{v}\in V$ (existence of zero vector)

④ For each vector $\vec{v}\in V$, $\exists\ -\vec{v}\in V$ s.t $\vec{v}+(-\vec{v})=\vec{0}$ (existence of vector negatives)

arithmetic properties:

$\forall\ \vec{u},\vec{v},\vec{w}\in V$ & $\forall\ c,d\in\mathbb{R}$

⑤ $\vec{u}+\vec{v}=\vec{v}+\vec{u}$ (vector addition is commutative)

⑥ $\vec{u}+(\vec{v}+\vec{w})=(\vec{u}+\vec{v})+\vec{w}$ (vector addition is associative)

⑦ $c\cdot(\vec{u}+\vec{v})=c\vec{u}+c\vec{v}$ (scalar multiplication distributes over vector addition)

⑧ $(c+d)\cdot\vec{u}=c\vec{u}+d\vec{u}$ (addition of scalars distributes over scalar multiplication)

⑨ $c(d\vec{u})=(cd)\vec{u}$ (multiplication of scalars is compatible with scalar multiplication)

⑩ $1\cdot\vec{u}=\vec{u}$ (unity law)

Function spaces

$\mathcal{F}$: set of all real-valued functions with domain $(-\infty,\infty)$

$\mathcal{F}=\{f\mid f:\mathbb{R}\to\mathbb{R}\}$

for $f,g\in\mathcal{F}$, we have $f=g$ if & only if $f(x)=g(x)$ $\forall x\in(-\infty,\infty)$

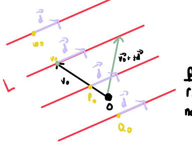
Trivial vector space: $V=\{0\}$

2. Lines & Planes

Parametric equation

$L=\{\vec{v}_0+t\vec{d}:t\in\mathbb{R}\}$

need: point & direction vector



Parallel

direction vectors are multiples.

Skew lines

only in $\mathbb{R}^3$, not parallel, don't intersect

Parametric form

$r=r_0+s\cdot v+t\cdot v$

need: one point, 2 independent directions, 2 parameters

Cross product

only in $\mathbb{R}^3$

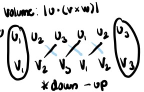
produces vector perpendicular to both

magnitude: $\|u\times v\|=\|u\|\cdot\|v\|\cdot\sin\theta$

area of parallelogram: $\|u\times v\|$

area of triangle: $\frac{1}{2}\|u\times v\|$

volume: $|u\cdot(v\times w)|$



u_1 u_2 u_3 u_1 u_2 u_3
v_1 v_2 v_3 v_1 v_2 v_3

*down - up
blue - black

Intersect

solve for parameters

Cartesian form

$ax+by+cz=d$

normal vector: (a,b,c)

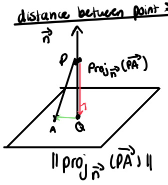
Plane through 3 points

1. find 2 direction vectors
2. compute $\vec{n}=d_1\times d_2$ → cross product
3. Use: $\vec{n}\cdot(x-x_0)=0$

   $d=\vec{n}\cdot v_0$

   $ax+by+cz=d$

distance between point & plane



$\|\text{proj}_{\vec{n}}(\vec{PA})\|$

intersections

intersection between line & plane

↳ reduce line to (x,y,z) and substitute them in the plane

intersection between planes

↳ eliminate 2 variables (separately from eachother)

4. Subspaces

Let $U\subseteq V$

as subset is a subspace if:

- contains zero vector
- closed under addition
- closed under scalar multiplication

5. Span

span = all linear combinations

↳ $\text{span}(v_1,\dots,v_n)$

*span is always a subspace, spanning set generates every vector

Then the set of vectors $\{v_1,\dots,v_m\}$ is called a spanning set for S. We also say that "S is spanned by the vectors $v_1,\dots,v_m$". Or sometimes say "the vectors $v_1,\dots,v_m$ span S".

6. Linear dependence

Dependent:

$\exists\ a_1v_1+\dots+a_nv_n=0$ where not all $a_i=0$

Independent:

only trivial solution

Row echelon form (REF)

1. All rows consisting entirely are at the bottom of the matrix.
2. The leading entry (pivot) of each nonzero row is to the right of the leading entry in the row above it.
3. every entry below each pivot is zero.

Reduced row echelon form (RREF)

satisfies all REF conditions &:

4. each pivot is equal to 1
5. each pivot is the only nonzero entry in its column

8. Basis & Dimension

Basis:

- lin independent
- spans the spaces

Dimension:

- number of vectors in a basis
- every basis has the same size.

9. Dimension theorems

every LI set $\le$ every spanning set

every spanning set can shrink to a basis

every LI set can grow to a basis

* 2 linear systems ar equivalent if they have the same general solution.

* every matrix is now equivalent to a unique matrix in RREF

1. If matrix $[A'|b']$ contains a row

   $[0\ 0\cdots 0|b]$ where $b\ne 0$

   it is inconsistent no solutions

2. otherwise it is consistent with a unique solution or infinite solutions

   ↳ every column has a leading 1 — ↳ at least one column doesn't have a leading 1

* if $\text{rank}([A|\vec{b}])>\text{rank}(A)$ system is inconsistent

* if $\text{rank}([A|\vec{b}])=\text{rank}(A)=$ # unknowns, unique solution

* if $\text{rank}([A|\vec{b}])=\text{rank}(A)<$ # unknowns, infinite solutions

10. Coordinates

If $B=\{v_1,\dots,v_n\}$ is a basis, then $x=a_1v_1+\dots+a_nv_n$

coordinates are $(a_1,\dots,a_n)$

11. Solving systems

$[A|b]$

use row operations:

1. swap
2. multiply row
3. add multiple of row

Possibilities: unique, infinite, no solution.

12. Rank

Rank = number of pivots

= number of nonzero rows in REF

= dimension of $\text{col}(A)$

= dimension of $\text{row}(A)$

Nullity = # of free variables

rank theorem: rank + nullity = n

13. Consistency

inconsistent if $[0\ 0\ 0\ 1|1]$ apears

otherwise consistent

14. Homogenous systems

$Ax=0$

- always consistent
- at least one solution (the zero vector)

15. applications

know how to

- determine flow networks
- solve systems
- interpret solutions

Theorem 16.16.

Let A be an $n\times n$ matrix. Then the following statements are all equivalent (either they are all true or they are all false):

1. $\text{rank}(A)=n$
2. $Ax=0$ has only the trivial solution
3. $Ax=b$ is consistent for all $b\in\mathbb{R}$
4. Every linear system $Ax=b$ has a unique solution.
5. The RREF of A is I_n.
6. $\text{null}(A)=\{0\}$.
7. $\text{col}(A)=\mathbb{R}^n$
8. $\text{row}(A)=\mathbb{R}^n$
9. $\text{rank}(A^T)=n$
10. The columns of A are L.I.
11. The rows of A are L.I.
12. The columns of A span $\mathbb{R}^n$.
13. The rows of A (transposed) span $\mathbb{R}^n$.
14. The columns of A form a basis for $\mathbb{R}^n$.
15. The rows of A form a basis for $\mathbb{R}^n$.
16. A is invertible.
17. A^T is invertible.

16. Matrix multiplication

Dimensions

$A_{m\times n}\cdot B_{n\times p}=C_{m\times p}$ (possible if cols of A = rows of B)

Properties

associative: $A(BC)=(AB)C$

distributive: $A(B+C)=AB+AC$

not commutative $AB\ne BA$

transpose: $(AB)^T=B^T\cdot A^T$

17. four fundamental subspaces

column space

span of columns

lives in: $\mathbb{R}^n$

dimension = rank

Row space

span of rows

dimension = rank

Null space

$Ax=0$

solutions live in $\mathbb{R}^n$

Left null space

solutions of

$A^Ty=0$

18. equivalent statements

square matrix A

following are equivalent:

- invertible
- $\det(A)\ne 0$
- full rank
- pivots in every row
- pivots in every column
- unique solution
- columns are LI
- columns span
- $\text{null}(A)=0$

* if one is true, all are true. if one is false, all are false

19. matrix inverse

exists if $\det(A)\ne 0$

finding inverse

argument: $[A|I]$

↓

row reduce

↓

finish $[I|A^{-1}]$

Properties

$AA^{-1}=I$

$(A^{-1})^{-1}=A$

$(AB)^{-1}=B^{-1}A^{-1}$

$(A^T)^{-1}=(A^{-1})^T$

20. orthogonality

orthogonal → dot product zero

orthonormal → orthogonal & length one

Projection: $\text{proj}_u(v)=\dfrac{v\cdot u}{u\cdot u}\cdot u$

distance: $v-\text{proj}_u(v)$

21. Gram-Schmidt

Given: $v_1,v_2,\dots$

construct: $u_1=v_1$

then: $u_2=v_2-\text{proj}_{u_1}(v_2)$

then: $u_3=v_3-\text{proj}_{u_1}(v_3)-\text{proj}_{u_2}(v_3)$

normalize: $e_i=\dfrac{u_i}{\|u_i\|}$

22. orthogonal complement

Definition: all vectors orthogonal to W.

Notation: $W^\perp$

Important theorem: $\dim(W)+\dim(W^\perp)=n$

23. Determinants

2x2

$ad-bc$ $\begin{bmatrix}a&b\\c&d\end{bmatrix}$

↳ $\det(\text{Matrix}_{2\times 2})$

3x3

cofactor expansion

The column space, the row space, & the null space of a matrix

column space of A: $\text{col}(A)$

$\text{Col}(A)=\text{span}\{\vec{c_1},\dots,\vec{c_n}\}$

row space of A: $\text{row}(A)$

$\text{row}(A)=\text{span}\{\vec{r_1},\dots,\vec{r_m}\}$

null space of A: $\text{null}(A)$

$\text{null}(A)=\{\vec{x}\in\mathbb{R}^n:\underset{m\times n}{A}\underset{n\times 1}{\vec{x}}=\underset{m\times 1}{\vec{0}}\}$

$m\times n$ matrix

$c_1,\dots,c_n$ (columns)

$r_1,\dots,r_m$ (rows)

Row space algorithm

$\dim(\text{row}(A))=\text{rank}(A)$.

column space algorithm

$\dim(\text{col}(A))=\text{rank}(A)$

Theorem 15.15

1. $\text{rank}(A)=n=$ # rows = # columns.
2. $\text{rank}(A^T)=n=$ # rows = # columns.
3. every linear system $Ax=b$ has a unique solution.
4. The RREF of A is I_n (identity matrix).
5. $\text{null}(A)=\{0\}$ (trivial vector space).
6. $\text{col}(A)=\mathbb{R}^n$.
7. $\text{row}(A)=\mathbb{R}^n$.
8. The columns of A are L.I.
9. The rows of A are L.I.
10. the columns of A form a basis for $\mathbb{R}^n$.
11. The rows of A form a basis for $\mathbb{R}^n$.

*Let A be an $n\times n$ matrix Then the following are equivalent

matrix multiplication

$$\begin{bmatrix}a&b\\c&d\end{bmatrix}_{2\times 2}\cdot\begin{bmatrix}e&f\\g&h\end{bmatrix}_{2\times 2}=\begin{bmatrix}ae+bg&af+bh\\ce+dg&fc+dh\end{bmatrix}_{2\times 2}$$

$$\begin{bmatrix}a&b\\c&d\end{bmatrix}_{2\times 2}\cdot\begin{bmatrix}e&f&g\\h&i&j\end{bmatrix}_{2\times 3}=\begin{bmatrix}ae+bh&af+bi&ag+bj\\ce+dh&cf+di&cg+dj\end{bmatrix}_{2\times 3}$$

$k(AB)=(kA)B=A(kB)$

$(AB)^T=B^TA^T$

$AI=A$ & $IB=B$

$A0=0$ $0B=0$

18. Determinants

Def: let A be an n×n matrix. The determinant of A, denoted det(A), is defined as follows:

- If n=1, then A=[a] & define det(A) as: det(A)=a
- If n=2, then $A=\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ & we define det(A)=ad-bc
- If n=3, then $A=\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$ & we define det(A) as
  $\hookrightarrow (-1)^{1+1}a\begin{vmatrix} e & f \\ h & i \end{vmatrix} + (-1)^{1+2}b\begin{vmatrix} d & f \\ g & i \end{vmatrix} + (-1)^{1+3}c\begin{vmatrix} d & e \\ g & h \end{vmatrix}$
- If n>3 & $A=\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & & \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$, then we define det(A) as $\det(A)=a_{11}\cdot\det(A_{11})-a_{12}\cdot\det(A_{12})+\cdots+(-1)^{1+n}\det(A_{1n})$

↗ this way

* computing det(A) is called a cofactor expansion along the first row.

ex: compute det(A) where

$$A=\begin{bmatrix} 2 & 3 & 4 & 5 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

cofactor expansion along any row or column

$A=\begin{bmatrix} a_{11} & \cdots & a_{1n} \\ & & \\ a_{n1} & \cdots & a_{nn} \end{bmatrix}$ Then det(A) can be calculated using

- The cofactor expansion along row i (for any i): $\det(A)=(-1)^{i+1}a_{i1}\det(A_{i1})+(-1)^{i+2}a_{i2}\det(A_{i2})+\cdots+(-1)^{i+n}a_{in}\det(A_{in})$.
- The cofactor expansion along column j (for any j): $\det(A)=(-1)^{1+j}a_{1j}\det(A_{1j})+(-1)^{2+j}a_{2j}\det(A_{2j})+\cdots+(-1)^{n+j}a_{nj}\det(A_{nj})$

Quick properties of the determinant

1. If A has a row or column of zero, then det(A)=0
2. $\det(A)=\det(A^T)$

n×n matrix is called triangular if $(A)_{i,j}=0$ whenever $j>i$

$$\begin{bmatrix} a_{11} & \cdots & & a_{1n} \\ 0 & a_{22} & \cdots & \vdots \\ 0 & 0 & a_{33} & \vdots \\ \vdots & \vdots & \vdots & a_{nn} \\ 0 & 0 & 0 & \end{bmatrix}$$

* the determinant of a triangular matrix is the product of its diagonal entries.

Properties of determinant

A & B are n×n matrices

1. $\det(kA)=k^n\det(A)$
2. $\det(AB)=\det(A)\det(B)$
3. $\det(A)=0 \iff$ A is not invertible
4. If A is invertible, then $\det(A^{-1})=\frac{1}{\det(A)}$

Theorem 18.16. [A LONG(ER) LIST OF EQUIVALENT STATEMENTS FOR SQUARE MATRICES!]

Let A be an $n\times n$ matrix.

Then the following statements are all equivalent (either they are all true or they are all false):

1. $\mathrm{rank}(A)=n$.
2. $A\mathbf{x}=\mathbf{0}$ has only the trivial solution.
3. $A\mathbf{x}=\mathbf{b}$ is consistent for all $\mathbf{b}\in\mathbb{R}^n$.
4. Every linear system $A\mathbf{x}=\mathbf{b}$ has a unique solution.
5. The RREF of A is I_n.
6. $\mathrm{null}(A)=\{\mathbf{0}\}$.
7. $\mathrm{col}(A)=\mathbb{R}^n$.
8. $\mathrm{row}(A)=\mathbb{R}^n$.
9. $\mathrm{rank}(A^T)=n$.
10. The columns of A are linearly independent.
11. The rows of A are linearly independent.
12. The columns of A span $\mathbb{R}^n$.
13. The rows of A (transposed) span $\mathbb{R}^n$.
14. The columns of A form a basis for $\mathbb{R}^n$.
15. The rows of A form a basis for $\mathbb{R}^n$.
16. A is invertible.
17. A^T is invertible.
18. $\det(A)\neq 0$.

Theorem 18.11. [EFFECTS OF ROW/COLUMN OPERATIONS ON DETERMINANT] Let A be an $n\times n$ matrix. Let B be obtained from A by performing one elementary row (or column) operation. Then

$\boxed{R_i \leftrightarrow R_j}$ If B is obtained from A by interchanging two rows, then $\det(B)=-\det(A)$.

$\boxed{kR_i \to R_i}$ If B is obtained from A by scalar-multiplying one row by k, then $\det(B)=k\det(A)$.

$\boxed{R_i + kR_j \to R_i}$ If B is obtained from A by adding a multiple of one row to another row, then

$$\det(B)=\det(A).$$

$\boxed{C_i \leftrightarrow C_j}$ If B is obtained from A by interchanging two columns, then $\det(B)=-\det(A)$.

$\boxed{kC_i \to C_i}$ If B is obtained from A by scalar-multiplying one column by k, then $\det(B)=k\det(A)$.

$\boxed{C_i + kC_j \to C_i}$ If B is obtained from A by adding a multiple of one column to another column, then

$$\det(B)=\det(A).$$

17. Orthogonality, Gram-Schmidt algorithm & orthogonal complements

Dot product & orthogonal vectors

$U=(U_1,\dots,U_n)$, $V=(V_1,\dots,V_n)$

$\vec{U}\cdot\vec{V}=0$

Observations:

- Any orthogonal set of vectors in $\mathbb{R}^n$ has at most n vectors.
- Any orthogonal set of n vectors in $\mathbb{R}^n$ form a basis of $\mathbb{R}^n$
- Every orthogonal set is LI, but not every LI set is orthogonal

orthogonal projection

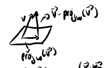
$$w=\overset{\rightarrow \mathrm{proj}_{\vec{w}_1}(\vec{w})}{\left(\frac{w\cdot w_1}{\|w_1\|^2}\right)w_1}+\overset{\rightarrow \mathrm{proj}_{\vec{w}_2}(\vec{w})}{\left(\frac{w\cdot w_2}{\|w_2\|^2}\right)\cdot w_2}+\cdots+\overset{\rightarrow \mathrm{proj}_{\vec{w}_m}(\vec{w})}{\left(\frac{w\cdot w_m}{\|w_m\|^2}\right)\cdot w_m}$$

↳ we can use the idea for a vector that doesn't belong to W.

Def 17.8.

W subspace of $\mathbb{R}^n$, $B=\{w_1,\dots,w_m\}$ be an orthogonal basis for W, let $V\in\mathbb{R}^n$

orthogonal projection of v onto the subspace W, denoted $\mathrm{proj}_W(V)$, is the vector


v
$\vec{v}-\mathrm{proj}_W(\vec{v})$
$\mathrm{proj}_W(\vec{v})$

$$\mathrm{proj}_W(\vec{v})=\left(\frac{\vec{v}\cdot\vec{w_1}}{\|\vec{w_1}\|^2}\right)\vec{w_1}+\cdots+\left(\frac{\vec{v}\cdot\vec{w_m}}{\|\vec{w_m}\|^2}\right)\vec{w_m}$$

* basis B must be orthogonal

Orthogonal complement

Def: Let U be a subspace of $\mathbb{R}^n$. The orthogonal complement of U, denoted $U^\perp$, is the set

$$U^\perp=\{\vec{v}\in\mathbb{R}^n : \vec{v}\cdot\vec{u}=0 \text{ for all } \vec{u}\in U\}$$

Theorem: (properties of orthogonal complement) Let U be a subspace of $\mathbb{R}^n$ Then:

1. $U^\perp$ is a subspace of $\mathbb{R}^n$
2. If $v\cdot u_i=0$ for each vector u_i in a spanning set $S=\{u_1,\dots,u_m\}$ of U, then $v\in U^\perp$.
3. for any matrix A, $(\mathrm{row}(A))^\perp=\mathrm{null}(A)$
4. $\dim(U)+\dim(U^\perp)=n$
5. $(U^\perp)^\perp=U$

Orthogonal sets

if u & v are orthogonal, we denote this as the set set of vectors $S=\{v_1,\dots,v_m\}$ is called orthogonal if the vectors are all orthogonal to eachother

Orthogonal bases:

a set $B=\{v_1,\dots,v_n\}$ of n vectors in $\mathbb{R}^n$ is called an O.B. for $\mathbb{R}^n$ if B is orthogonal.

Theorem 17.3.

Any orthogonal set linearly independent

Theorem 17.6. (expansion theorem: coord. to an orthogonal basis)

$B=\{w_1,\dots,w_m\}$ is an orthogonal basis for a subspace W of $\mathbb{R}^n$. Let $w\in W$ be an arbitrary vector in W Then $w=\left(\frac{w\cdot w_1}{\|w_1\|^2}\right)w_1+\left(\frac{w\cdot w_2}{\|w_2\|^2}\right)\cdot w_2+\cdots+\left(\frac{w\cdot w_m}{\|w_m\|^2}\right)\cdot w_m$

hence the coordinates of w relative to the ordered basis B are $\left(\frac{w\cdot w_1}{\|w_1\|^2},\dots,\frac{w\cdot w_m}{\|w_m\|^2}\right)$

Theorem 17.10. Best approximation theorems

Let W be a subspace of $\mathbb{R}^n$ & let $V\in\mathbb{R}^n$. then

1. $\mathrm{proj}_W(V)\in W$
2. $V-\mathrm{proj}_W(V)$ is orthogonal to every vector in W
3. $\mathrm{proj}_W(V)$ is best approximation to V out of all the vectors in W
4. $\mathrm{proj}_W(V)$ is the only vector in $\mathbb{R}^n$ that satisfies conditions 1 & above.

Gram-Shmidt algorithm

Let $W=\mathrm{span}\{U_1,\dots,U_m\}$. Let's create a new set $S=\{w_1,\dots,w_m\}$ as follows:

- Let $w_1=U_1$
- Let $w_2=U_2-\mathrm{proj}_{w_1}(U_2)$
- Let $w_3=U_3-\mathrm{proj}_{w_1}(U_3)-\mathrm{proj}_{w_2}(U_3)$
- Let $w_4=U_4-\mathrm{proj}_{w_1}(U_4)-\mathrm{proj}_{w_2}(U_4)-\mathrm{proj}_{w_3}(U_4)$

$\vdots$

- Let $w_i=U_i-\sum_{j=1}^{i-1}\mathrm{proj}_{w_j}(U_i)$
- Let $w_m=U_m-\sum_{i=1}^{m-1}\mathrm{proj}_{w_i}(U_m)$

next remove any zero vectors that are in S to obtain $B=\{w_i\in S : w_i\neq 0\}$. Then B is an orthogonal basis for W!

19. Eigenvalues & Eigenvectors

Def: Let A be an n×n matrix. Let $x\in\mathbb{R}^n$ be a nonzero vector & let $\lambda\in\mathbb{R}$ be a scalar.

- The vector x is called an eigenvector of A if $A\vec{x}=\lambda\vec{x}$ ↳ eigen vector ↳ eigen value
  - In this case we say that λ is an eigenvalue of A & we say that x is the corresponding eigenvalue of x.

finding eigenvalues: $\det(A-\lambda I)=0$

Def: Let A be a n×n matrix

- The characteristic polynomial of A is $\det(A-\lambda I)$
- Note: $\det(A-\lambda I)$ is a polynomial in unknown λ of degree n
- The roots of the characterstic polynomial of A are eigenvalues of A
- The algebraic multiplicity of an eigenvalue λ of A is the multiplicity of λ as a root of the characteristic polynomial of A.

LIMITATIONS ON DIAGONALIZABILITY

There are two main obstacles when it comes to diagonalizability:

- There might not be enough LI eigenvectors to form an eigen-basis. When can this happen?

  If the characteristic polynomial has a repeated root (alg. mult. >1), then it *can happen* that $\dim E_\lambda <$(alg. mult. λ). In this case, we won't be able to find n LI eigenvectors.

- Some of the eigenvalues could be complex numbers.

  In this case, we cannot find an eigen-basis of $\mathbb{R}^n$, so A is not diagonalizable "over $\mathbb{R}$".

Def: Let λ be an eigenvalue of an n×n matrix A.

- The λ-Eigenspace of A is the subspace:

$$\begin{aligned} E_\lambda &= \mathrm{null}(A-\lambda I) \\ &= \{\vec{x}\in\mathbb{R}^n : (A-\lambda I)\vec{x}=\vec{0}\} \\ &= \{\vec{x}\in\mathbb{R}^n : A\vec{x}=\lambda\vec{x}\} \end{aligned}$$

- The nonzero vectors in E_λ are the eigenvectors of A corresponding to λ.
- The geometrix multiplicity of λ is the dimension of the eigenspace E_λ.

Theorem: (Independence of eigenvectors for distinct eigenvalues) Let A be an n×n matrix. Suppose $\lambda_1,\lambda_2,\dots,\lambda_k$ are k distinct eigenvalues of A. If, for each $i=1,\dots,k$, the vector x_i is a λ-eigenvector of A, then the set $\{v_1,\dots,v_k\}$ is linearly independent (LI).

DIAGONALIZABILITY: AN EIGEN-BASIS FOR $\mathbb{R}^n$

Let A be an $n\times n$ matrix.

- The characteristic polynomial of A is a polynomial of degree n so it has at most n distinct real roots.
- Each root of the characteristic polynomial is an eigenvalue of A.
- Each eigenvalue of A has an eigenspace of dimension ≥ 1.
- In fact, for every eigenvalue λ of A, we have:

  $1\leq$ [geometric multiplicity of λ] $=\dim E_\lambda\leq$ [algebraic multiplicity of λ] (theorem!)

- Eigenvectors behave so simply with respect to multiplication by A that it would be wonderful to find a basis of $\mathbb{R}^n$ consisting entirely of eigenvectors of A. If we find such a basis, we'll call it an **EIGEN-BASIS FOR $\mathbb{R}^n$**.
- If we can find an eigen-basis for $\mathbb{R}^n$ consisting of eigenvectors of A, then we say that the matrix A is **DIAGONALIZABLE**.
- That means, given A, we are hoping to find n linearly independent eigenvectors for A.

20. Diagonalizability

def: Let A be an n×n matrix (with real entries). Then A is called diagonalizable (over $\mathbb{R}$) if there exists a basis of $\mathbb{R}^n$ consisting entirely of eigenvectors of A.

Why is this called "diagonizable"? Shouldn't the term be "eigenbasis-able"?

what's "diagonal" about "diagonalizable"?

Def: Let D be an n×n matrix Then we say that D is a diagonal matrix if

$$D=\begin{bmatrix} d_{11} & 0 & 0 & \cdots & 0 & 0 \\ 0 & d_{22} & 0 & \cdots & 0 & 0 \\ 0 & 0 & d_{33} & \cdots & 0 & \vdots \\ \vdots & \vdots & \vdots & \ddots & \vdots & \\ 0 & 0 & 0 & \cdots & d_{n-1,n-1} & 0 \\ 0 & 0 & 0 & \cdots & 0 & d_{nn} \end{bmatrix}$$

Theorem:

Let A be an n×n matrix

Then A is diagonalizable ⟺

HOW TO DIAGONALIZE A MATRIX

Suppose A is diagonalizable. How do we diagonalize it?

1. Find the eigenvalues of A which are the roots of its characteristic polynomial $\det(A-\lambda I)$.
   - ○ If A has any non-real eigenvalues (i.e. A has complex numbers with nonzero imaginary part), then, in MAT1341, STOP! A is not diagonalizable over $\mathbb{R}$ (although A might be diagonalizable over $\mathbb{C}$).
   - ○ If all eigenvalues of A are real, continue to the next step.
2. For each eigenvalue λ of A, find a basis for the λ-eigenspace $E_\lambda=\mathrm{null}(A-\lambda I)$.
3. Put all eigenspace basis vectors together.
   - ○ If the set has $<n$ vectors, STOP! A is not diagonalizable.

     this happens $\iff$ (geom. mult. of λ) $<$ (alg. mult. of λ) for at least one of its eigenvalues λ
   - ○ If the set has n vectors in it, then its an eigen-basis for $\mathbb{R}^n$ and A is diagonalizable!
4. (only if A is diagonalizable)
   - ○ Create the $n\times n$ matrix P whose columns are the eigen-basis you found in Step 3.
   - ○ Create the $n\times n$ diagonal matrix D whose diagonal entries are the eigenvalues of A (repeated up to their algebraic multiplicity and sorted in the same order as you put the eigen-basis columns into P).
5. Now $A=PDP^{-1}$ so we have "diagonalized" A!!!

* suppose A is diagonalizable & we'd like to compute "big" powers of A (many applications

$$A^t=(PDP^{-1})^t=PDP^{-1}\cdot PDP^{-1}\cdot PDP^{-1}\cdot\ \cdots\ \cdot PDP^{-1}\cdot PDP^{-1}=PD^tP^{-1}$$